

Predicting Hospital Readmission Rates for Diabetic Patients

Overview

- Submission Format: A zip file containing a PDF file of the written report and a Jupyter Notebook .ipynb file of the code

Please note that this is a group assignment:

Please arrange yourself into groups and then register your group with the Teaching Assistants.

Assignment Description

One of the common application areas for machine learning is in healthcare. Many clinical decisions require accumulation of information from a large array of sources to make a diagnosis or decide patient treatment. Such problems are well-suited to machine learning.

This assignment asks you to predict readmission for diabetic patients admitted to hospital, i.e., what is the likelihood that a patient will need to come back to hospital given the data about this visit. This is based on a database of 101766 hospital visits with a set of 50 features covering patient information, treatment and prescribed medication during the hospital stay.

The database covers patients admitted to 130 hospitals in the US over a period of 10 years to 2008. The original data came from this source:

<https://archive.ics.uci.edu/dataset/296/diabetes+130-us+hospitals+for+years+1999-2008>

You will need to build a model that predicts the readmission status of the patient. There are three categories: No readmission; readmission after 30 days and readmission in less than 30 days.

You can use the training data along with a suitable evaluation method (e.g. splitting the training data into training and validation sets) to train and validate your own models. You will be assessed primarily not so much on the final predictive accuracy that you obtain, but rather on your approach when attempting this task, and on the level of understanding that you demonstrate. The marking schedule describes the criteria you will be assessed on.

This is not an easy dataset to work with. There are missing values that you will need to account for. The prediction itself is difficult and I do not expect miraculous results. I just want to see that you have made a decent attempt to tackle this problem and shown understanding of machine learning in the methods we have covered.

Be creative and ask questions of the data. Think about methods to generate features missing in the test data. Most of all, please reflect upon what you have learned during the term and seek to apply machine learning in a way that is appropriate for this task. You can use any of the algorithm implementation of the methods we covered in the course, in sklearn, Tensorflow or from any other source you may find.

Be sure that all members of the group participate. There are plenty of tasks to be done. Your contributions will be peer-assessed.

Data Description

The data consists of the following files:

- diabetic_data_training.csv (training data with 91589 instances)
- diabetic_data_test.csv (test data with 10177 instances)
- FeatureTable.csv (Descriptions of the features)
- IDS_mapping.csv (The meaning of the numeric IDs for some features)

Do not amend or recreate the test set. This is separate from the training set and must not be used in model selection. I would like all teams to evaluate with the same data. The test set should be used for final evaluation only.

The assignment submission will take the form of:

1. A PDF file containing a 7-page report of your results on the task.
2. A Jupyter notebook containing the **Python** source code of your approach as well as (brief) in-line documentation

Submission Format & Structure

The assignment submission will take the form of a zip file containing:

1. A **PDF file** containing a **report** of the task (this should be at most 7 A4 sides in length, excluding references, appendix). This PDF should be prepared using the LATEX files included, which provide a format similar in style to "preprint" publications such as arXiv. For multiple people to work on the same report simultaneously, I recommend overleaf.com.

2. A **Jupyter notebook .ipynb file** containing the **Python source code** of your approach as well as (brief) in-line documentation. The notebook should include an analysis of the performance of your classifier on the data from the test set file.

The PDF report should adopt the following structure:

1. **Introduction**

A brief description of your approach to the problem and the results that you have obtained on the training data.

2. **Data Transformation & Exploration**

Any transformations that you apply to the data prior to training. Also, any exploration of the data that you performed such as visualization, feature selection, etc.

3. **Methodology Overview**

Start by describing in broad terms your methodology. Include any background reading you may have done and a step by step description of how you have trained and evaluated your model. Describe any feature engineering that you have applied. If you had attempted different approaches prior to landing on your final methodology, then describe those approaches here.

It is perfectly normal that you tried many methods and fail in many of them. But it is really the failures and the process of identifying why the model fails that demonstrates your understanding of the method.

4. **Model Training & Validation**

This contains a breakdown of how your model was trained and evaluated.

5. **Results**

Here you show the results that you obtain using your model on the training data. If you have multiple variations or approaches, this is where you compare them.

6. **Final Predictions on Test Set**

This is the section where you perform your final predictions on the test set using the model that you have trained in the previous section.

7. **Conclusion**

This is the section where you consider your findings and suggest avenues for future research.

References

List of relevant academic papers or sources used in your coursework and cited in the main report.

The Notebook should adopt the following structure:

1. Introduction

A brief precis of the equivalent section in your report.

2. Data Import

This section is how you import the data into the notebook. It should be written in such a way that I can modify it to run on my own machine by simply changing the location of the training data and any additional data sources that you have used.

3. Data Transformation & Exploration

Code for the equivalent section in your report, together with in-line documentation of that code.

4. Methodology Overview

Code for the equivalent section in your report, together with in-line documentation of that code.

5. Model Training & Validation

Code for the equivalent section in your report, together with in-line documentation of that code.

6. Results

Code for the equivalent section in your report, together with in-line documentation of that code.

7. Final Predictions on Test Set

Code for the equivalent section in your report, together with in-line documentation of that code.

Note:

- Your notebook need only contain brief in-line documentation, while the PDF should contain a more detailed description.
- You will be assessed primarily on the contents of your PDF report. The notebook is required so that we can check that your results are replicable.
- Keep in mind that your notebook should be written in such a way that we can modify the location of the data and then step through your notebook to obtain the same results as you have submitted.

Marking Guidelines

All reports will be marked against the marking criteria below.

The mark weighting for each section is as follows:

- **Methodology (15%)**

How well is the methodology described? How appropriate is it to the task at hand? Have any extra data sources been used and if so are they useful? Have you done more than just apply a classifier to the training data?

- **Evaluation Strategy (15%)**

Has a suitable evaluation strategy been used so as to avoid any possible bias? If your methodology contains multiple parameters, how have the final parameter values been chosen? Have you used any form of cross validation?

- **Presentation of Results (15%)**

Have you presented results on the training data? Are the results presented appropriate and displayed in an easy to interpret manner? Do they reveal any extra insights about how your model performs?

- **Interest of Approach (40%)**

How interesting and novel is your approach (regardless of predictive accuracy)? Have you used any extra data sources, or transformed the training data in an interesting way? Have you done something that is beyond simply using a standard classifier on the training data?

- **Format, structure, referencing, and clarity of writing/code (15%)**

Are your final notebook and report well laid out and does the write-up follow a clear structure? Have you included any references to show background research/reading? Is your writing free from spelling, punctuation, and grammatical errors and is your code well commented?

Q&A

1. Can we use methods that are not taught in the class?

The answer is no. You can only use the methods taught in the class.

Remember the purpose of the assessment is to test your understanding of the methods introduced in the module. They should be part of toolbox for data analytics from now on. As an engineer, you are expected to be familiar with the tools in the toolbox and use them adaptively in your task.

2. Is the coursework about choosing the best method among the ones introduced in the module?

Technically, this is for you to figure out but I will give a hint here. Just as an electrician who tries to fix your power supply or a plumber to fix your water supply, they first perform diagnosis to locate where the problem is, then normally use a series of combinations of tools to solve the problem. Your role is similar. So there might be no 'best' method to use but a creative combination of methods.

3. Are white-box methods preferred?

The general answer is yes, but not because they are superior in performance. They often are not. It is because the coursework is to show that you understand what we teach. The best way of showing it is:

1. Justify using a particular method to begin with.
2. Analyse the results and make guesses about what went wrong or what could be improved.
3. Do experiments to show whether your guesses are right.
4. If your guess is right, propose to use another method to address the identified shortcoming of the previously tried method.
5. Go to Step 2, until you need to stop trying new methods and start writing.

A white-box will make some of the steps above much easier.

Also, the above steps describe a common workflow of a data analyst.